

Exact Support Radius for Consecutive Vanishing Fourier Coefficients

A sharp consecutive-block result for Problem 44

July 20, 2026

Abstract

Let $1 \leq m \leq M$ be integers. We determine the infimal radius of the support of a nonzero smooth nonnegative function on the circle whose Fourier coefficients vanish at every frequency $m, m+1, \dots, M$. The answer is

$$r_{m,M} = \frac{M - m + 1}{2(M + m)}.$$

The lower bound follows from an explicit real trigonometric polynomial with spectrum contained in $\{\pm m, \dots, \pm M\}$ that is nonnegative on the extremal arc. Sharpness follows from an explicit positive atomic measure on $M - m + 2$ equally spaced points in that arc; convolving it with a small bump gives smooth examples at every larger radius. In particular, for the consecutive block $\{n, n+1, \dots, 2n\}$ highlighted in Problem 44 of Steinerberger's list, the exact infimum is $(n+1)/(6n)$, which tends to $1/6$.

1 The result

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and use the convention

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x) e^{-2\pi i k x} dx.$$

For integers $1 \leq m \leq M$, define

$$r_{m,M} = \inf \left\{ r > 0 : \text{there exists } 0 \not\equiv f \in C^\infty(\mathbb{T}), \quad f \geq 0, \right. \\ \left. \text{supp } f \subset (-r, r), \quad \widehat{f}(k) = 0 \quad (m \leq k \leq M) \right\}.$$

The interval is interpreted inside the standard representative $(-1/2, 1/2)$ of the circle.

Theorem 1. *For every $1 \leq m \leq M$,*

$$r_{m,M} = \frac{M - m + 1}{2(M + m)}.$$

The infimum is sharp in the following sense: a positive atomic probability measure exists on the closed extremal interval, and for every $r > r_{m,M}$ a smooth nonnegative probability density exists with support in $(-r, r)$ and all the prescribed Fourier coefficients equal to zero.

Two cases are worth recording immediately.

Corollary 1. For $n \geq 1$,

$$r_{1,n} = \frac{n}{2(n+1)}, \quad r_{n,2n} = \frac{n+1}{6n}.$$

Thus annihilating the first n frequencies requires support on nearly the whole circle, while annihilating the block from n through $2n$ still requires an interval of asymptotically fixed radius $1/6$.

We prove the theorem in angular coordinates $\theta = 2\pi x$. Put

$$L = M - m, \quad D = M + m, \quad \alpha = \frac{\pi(L+1)}{D}.$$

The claimed radius is exactly $\alpha/(2\pi)$.

2 A one-sign dual certificate

Partition the extremal arc $[-\alpha, \alpha]$ by

$$\theta_j = -\alpha + \frac{2\pi j}{D}, \quad 0 \leq j \leq L+1. \quad (1)$$

These are $L+2$ equally spaced points. Using the L internal points, define

$$F(\theta) = e^{im\theta} \prod_{j=1}^L (e^{i\theta} - e^{i\theta_j}), \quad T(\theta) = (-1)^L \operatorname{Re} F(\theta), \quad (2)$$

where an empty product is 1.

Lemma 1 (One-sign certificate). *The real trigonometric polynomial T has Fourier spectrum contained in $\{\pm m, \dots, \pm M\}$. Moreover,*

$$T(\theta) \geq 0 \quad (-\alpha \leq \theta \leq \alpha),$$

and its zeros on this interval are exactly the points $\theta_0, \dots, \theta_{L+1}$.

Proof. The polynomial F is a linear combination of $e^{im\theta}, e^{i(m+1)\theta}, \dots, e^{iM\theta}$, proving the spectral claim. The internal nodes satisfy $\sum_{j=1}^L \theta_j = 0$, so

$$F(\theta) = (2i)^L e^{iD\theta/2} \prod_{j=1}^L \sin\left(\frac{\theta - \theta_j}{2}\right). \quad (3)$$

Fix $0 \leq j \leq L$ and let $\theta_j < \theta < \theta_{j+1}$. Write

$$u = \frac{D}{2}(\theta - \theta_j) \in (0, \pi).$$

The real product in (3) has sign $(-1)^{L-j}$, while

$$e^{iD\theta/2} = e^{-i\pi(L+1)/2} (-1)^j e^{iu}.$$

Consequently

$$(-1)^L \operatorname{Re} F(\theta) = 2^L \left| \prod_{h=1}^L \sin\left(\frac{\theta - \theta_h}{2}\right) \right| \sin u > 0.$$

It vanishes at every partition point and nowhere else on the arc. □

Proof of the lower bound in Theorem 1. Suppose $r < \alpha/(2\pi)$ and that f satisfied the defining conditions. In angular coordinates, its support lies strictly inside $[-\alpha, \alpha]$. Because the spectrum of T consists only of the forbidden frequencies and their negatives,

$$\int_{-\pi}^{\pi} f\left(\frac{\theta}{2\pi}\right) T(\theta) d\theta = 0.$$

On the other hand, Lemma 1 says that the integrand is nonnegative and that T is strictly positive except at finitely many points. A nonzero nonnegative L^1 function cannot be supported on a finite set, so the integral must be positive, a contradiction. $\square$

3 A positive extremal measure

We now use all the nodes in (1). Set

$$z_j = e^{i\theta_j}, \quad c_j = \left(\prod_{\substack{0 \leq h \leq L+1 \\ h \neq j}} |z_j - z_h| \right)^{-1} > 0, \quad w_j = \frac{c_j}{\sum_{h=0}^{L+1} c_h}.$$

Lemma 2 (Positive quadrature). *The probability measure*

$$\mu = \sum_{j=0}^{L+1} w_j \delta_{\theta_j}$$

satisfies

$$\int e^{ik\theta} d\mu(\theta) = 0 \quad (m \leq k \leq M).$$

Proof. Let

$$P(z) = \prod_{h=0}^{L+1} (z - z_h), \quad \lambda_j = \frac{1}{P'(z_j)}.$$

Lagrange interpolation, applied to $1, z, \dots, z^L$ and followed by comparison of the z^{L+1} coefficient, gives

$$\sum_{j=0}^{L+1} \lambda_j z_j^q = 0, \quad 0 \leq q \leq L. \quad (4)$$

We claim that the complex numbers $z_j^{-m} \lambda_j$ all lie on the same open ray. Indeed, since the nodes are ordered around an arc and have sum of arguments zero,

$$\frac{P'(z_j)}{|P'(z_j)|} = i^{L+1} (-1)^{L+1-j} e^{iL\theta_j/2}.$$

Using $D = 2m + L$ and $D\theta_j/2 = -\pi(L+1)/2 + \pi j$, the phase of $z_j^{-m}/P'(z_j)$ is independent of j . Its modulus is precisely c_j . Thus there is a single nonzero constant C such that

$$w_j = C z_j^{-m} \lambda_j$$

for all j , with C chosen to make the weights sum to 1.

If $k = m + q$ with $0 \leq q \leq L$, equation (4) now gives

$$\sum_{j=0}^{L+1} w_j z_j^k = C \sum_{j=0}^{L+1} \lambda_j z_j^q = 0.$$

□

Proof of sharpness in Theorem 1. Lemma 2 supplies a positive probability measure supported on the closed angular arc $[-\alpha, \alpha]$. Let ρ_ε be a nonnegative smooth probability density supported in $(-\varepsilon, \varepsilon)$ on the circle. The convolution $\mu * \rho_\varepsilon$ is smooth and nonnegative, and its Fourier coefficients are the products of the corresponding Fourier coefficients of μ and ρ_ε . Hence they still vanish for $m \leq k \leq M$. Its support lies in $(-\alpha - \varepsilon, \alpha + \varepsilon)$. Given any $r > \alpha/(2\pi)$, choosing $\varepsilon < 2\pi r - \alpha$ completes the construction. □

4 Consequences for Problem 44

For the block $A = \{n, n+1, \dots, 2n\}$, Theorem 1 gives

$$\inf r = \frac{n+1}{6n} = \frac{1}{6} + \frac{1}{6n}.$$

Thus the required support does not shrink with the frequency scale when a full octave of consecutive frequencies is removed. This is much stronger in this family than the candidate estimate

$$r \gtrsim \left(\sum_{a \in A} \frac{1}{a^2} \right)^{1/2} \asymp n^{-1/2}$$

suggested in the problem list.

At the other extreme, a singleton $A = \{m\}$ gives the exact value $r = 1/(4m)$, and an initial block $A = \{1, \dots, n\}$ gives $r = n/(2(n+1))$. The general finite-set problem remains open; the argument here exploits the uninterrupted frequency block in an essential way.

5 Novelty boundary

The proof uses classical ingredients from trigonometric moment problems—separating trigonometric polynomials, Lagrange interpolation, and positive quadrature. A targeted search did not locate the formula in Theorem 1 stated for this support-minimization problem, but the result may overlap older Chebyshev-system or truncated-moment literature. The mathematical claim is self-contained; priority should be checked independently before any novelty claim is made.

References

- [1] S. Steinerberger, *A Collection of Open Problems*, Problem 44, updated 2025. [online problem list](#).